

A Comprehensive Guide to User Rights Management

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Introduction

The Cisco IOS (Internetwork Operating System) provides admins with the means to manage and configure Cisco devices. While Cisco's command structure is proprietary, it exemplifies fundamental concepts in network device management that are applicable across various vendors. Similar command-line interfaces (CLIs) and configuration principles can be found in systems from other major networking companies like Aruba, Juniper and Broadcom, allowing network professionals to apply their skills across different platforms with some adaptation.

This article provides a cheat sheet that you can use as a quick command guide for managing and configuring Cisco network devices. Screenshots are provided to show some commands step by step in action; they were taken using a console connection to the router or switch.

Overview of Command-Line Interface (CLI) Structure

Command Modes

The Cisco IOS command-line interface (CLI) is structured hierarchically with the following command modes:

- **User EXEC mode** — This is the default mode when logging into a Cisco device. It offers limited access, allowing only basic monitoring commands and restricting the ability to make configuration changes. It is identifiable by a prompt ending with `>`.
- **Privileged EXEC mode** — This elevated mode grants access to all show commands and system-level operations. It provides complete visibility into the system but no configuration access. It also enables users to perform device diagnostics and file management tasks. This mode is denoted by a prompt ending with `#`.
- **Global Configuration mode** — This highly privileged mode allows you to make system-wide configuration changes and provides access to various sub modes for specific configurations, such as interface settings. In this mode, the prompt includes **(config)#**.

Moving Between Command Modes

When you first log into a Cisco router or switch, you will be in User EXEC mode. Use the following commands to switch to more privileged modes:

- **enable** — Used to access Privileged Exec mode
- **config terminal** (or simply **config t**) — Used to access Global Configuration mode

These commands are shown in the screenshot below:

```
Router>enable
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router (config) #
```

Global Configuration mode can be further divided into several sub-modes that permit you to configure different components. Here are those sub-modes and the commands to access them:

Interface Configuration mode	<code>interface [interface-type] [interface-number]</code>
Subinterface Configuration mode	<code>interface [interface-type] [interface-number].[subinterface-number]</code>
Router Configuration mode	<code>router [routing-protocol]</code>
Line Configuration mode	<code>line [line-type] [line-number]</code>

Now that you know the commands to move to the higher privileged modes, here are the commands to work your way back out of them.

- **exit** — Used to move back one level in the configuration mode hierarchy. The following screenshot shows the admin returning to Privileged Exec mode:

```
Router#enable
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router (config) #exit
Router#
```

- **end** — Used to immediately return to Privileged EXEC mode from any configuration mode. This is useful for quickly exiting nested configuration levels.

Displaying System Configuration

After logging onto a Cisco device, you probably want to view its basic system settings. This is easily done by typing **show running-config** or simply **show run**. This command displays the active configuration stored in the device's memory.

Notice in the screenshot below that we first tried the command while in User Exec mode, where it failed due to insufficient privileges. The screenshot shows only a portion of the running config.

```
Router>show run
Invalid input detected at
Router>enable
Router#show run
Building configuration...
marker.
Current configuration: 646 bytes
!
version 16.6.4
no service timestamps log datetime msec no service timestamps debug datetime
msec no service password-encryption
!
hostname Router
!
```

Other commands related to configuration include the following:

<code>show startup-config</code>	Displays the saved configuration stored in the device's NVRAM, which will be loaded when the device starts up
<code>copy running-config startup-config</code>	Replaces the startup config with the active config when the Cisco network device initializes
<code>copy startup-config running-config</code>	Merges the startup config with the currently active config in RAM
<code>write erase</code> <code>erase startup-config</code>	Deletes the startup config

Shortcuts and Command History

Here are a few additional commands for quick navigation:

- **Ctrl+Z** — Use this key combination at any time to instantly exit any configuration mode and return to Privileged EXEC mode
- **Tab key** — Use this key to auto-complete partial commands or show possible completion options.
- **Up & down arrow keys** — Use these keys to cycle through previous commands.
- **show history** — Use this command to display the command history buffer.

Device Setup and Basic Configuration

Renaming a Device

The running config displayed in the earlier screenshot showed router as the hostname. This is the default name for a Cisco router, just as **switch** is the default name for a Cisco switch. The device's name appears in the command prompt.

Use the **hostname** command to change a device's name. Select a unique name for each device to ensure easy identification within the network. In the example below, we changed the router's name to **NorthOfficeRouter**. Notice how the new name is a part of the final prompt.

```
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router (config)#hostname NorthOfficeRouter
NorthOffice Router (config)#
```

Assigning an IP Address

You will also want to assign an IP address to the device to remotely manage it. Choose the interface you want to assign the IP address to and complete the following steps:

1. Enter Interface Configuration mode:

```
interface <interface-name>
```

2. Assign an IP address and subnet mask:

```
ip address <IP-address>
```

3. Enable the interface:

```
no shutdown
```

Here is what this procedure looks like in the Cisco IOS. Note that config-if indicates that you are in Interface Configuration sub-mode.

```
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router (config) #hostname NorthOffice Router
NorthOffice Router (config)#
NorthOffice Router (config)#
NorthOfficeRouter (config)#interface GigabitEthernet0/0/1
NorthOfficeRouter (config-if)# ip address 192.168.1.1 255.255.255.0
NorthOfficeRouter (config-if)#no shutdown

NorthOfficeRouter (config-if)#
*LINK-5-CHANGED: Interface GigabitEthernet0/0/1, changed state to up

NorthOffice Router (config-if)#
```


Configuring Passwords

Because we logged onto a new device, we didn't have to enter a password. Here are the commands for enabling passwords to enhance security:

<code>enable password <pass-value></code>	Defines the password required when using the enable command
<code>enable secret <pass-value></code>	Sets the password required for any user to enter enable mode and encrypts it
<code>service password-encryption</code>	Merges the startup config with the currently active config in RAM

Configuring Banners

There are also a variety of banners you can configure:

- Message of the day (motd) banner:
`inbanner motd #Unauthorized access prohibited#`
- Login banner:
`banner login #Please enter your credentials#`
- Exec banner:
`banner exec #Welcome to the network#`

Switch Configuration

Switches serve a different purpose than routers and therefore utilize different commands. Here are some key concepts to understand:

- **Switchport mode** — The switchport mode determines how the port handles VLAN traffic. The three main modes are **access mode**, **trunk mode** and **dynamic mode**.
- **Duplex speed** — Duplex settings determine whether the port can send and receive data simultaneously. You can assign **half**, **full** or **auto**. **Auto** allows the switch to negotiate the duplex setting with the connected device.
- **Speed** — This setting determines the port's data transmission rate. Depending on the switch's capabilities, the speed can be set to 10 Mbps, 100 Mbps, 1000 Mbps (1 Gbps) or auto.

VLAN Configuration

VLAN configuration is a crucial aspect of network segmentation and management. In Global Configuration mode, you can use the following commands:

<code>vlan <vlan-id></code>	Creates a VLAN
<code>Vlan <vlan-name></code>	Assigns a name to the VLAN
<code>vlan switchport access vlan <vlan-id></code>	Sets the VLAN that the interface belongs to
<code>no vlan <vlan-id></code>	Deletes a VLAN
<code>show vlan</code>	Display the VLAN configuration

The screenshot below shows the commands for basic port configuration:

```
Switch(config)#vlan 10
Switch(config-vlan)#name wireless
Switch(config-vlan)#exit
Switch (config)#interface FastEthernet 0/1
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 10
Switch(config-if)#
```

Spanning Tree Protocol (STP) Commands

Spanning Tree Protocol is a feature of Cisco switches that can help prevent network loops. STP is enabled by default on Cisco switches for all VLANs. You can modify STP settings globally for the entire switch, or apply specific configurations to individual interfaces or VLANs for fine-tuned control over the STP behavior across different parts of your network. The command is as follows:

```
spanning-tree mode rapid-pvst
```

Configuring Trunk Ports

Ports that need to carry multiple VLAN traffic need to be trunked. You first need to trunk the port and then assign the VLANs. Here are the commands:

```
switchport trunk native vlan <vlan-id>
```

Configuring trunk ports on Cisco switches involves using the **switchport trunk** and **switchport trunk allowed vlan** commands. Here are more details on these commands:

<code>switchport trunk native vlan <vlan-id></code>	Enables trunk mode for the port and sets the native VLAN for untagged traffic on the trunk
<code>switchport trunk allowed vlan <vlan-id></code>	Adds the specified VLANs to the current list
<code>switchport trunk allowed vlan remove <vlan-id></code>	Removes the specified VLANs from the allowed list

The screenshot below shows these VLAN commands in action:

```
Switch#config t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config) #interface GigabitEthernet0/1
Switch(config-if)# switchport trunk native vlan 40
Switch(config-if)# switchport trunk allowed vlan 10
Switch(config-if)# switchport trunk allowed vlan 20,30
Switch(config-if)# switchport trunk allowed vlan remove 10
Switch(config-if)#
```

Networking Fundamentals

IP Addressing and Subnetting

The following commands are used for IP addressing and subnetting:

<code>ip address <ip-value> <bnet-value></code>	Assigns an IP and subnet mask
<code>show ip interface <interface-number></code>	Displays the status of a network interface as well as a detailed listing of its IP configurations and related characteristics
<code>show ip interface brief</code>	Provides a concise summary of the IP interface status and configuration
<code>ip address <ip-value> <subnet-value> secondary</code>	Assigns a secondary IP address
<code>no ip address</code>	Removes an IP address

The screenshot below shows the output from the show ip interface brief command:

NorthOfficeRouter#show	ip interface	brief				Protocol
Interface	IP-Address	OK?	Method	Status		down
GigabitEthernet0/0/0	unassigned	YES	unset	administratively	down	down
GigabitEthernet0/0/1	192.168.1.1	YES	manual	up		down
GigabitEthernet0/0/2	172.16.0.254	YES	manual	administratively	down	down
Vlan1	unassigned	YES	unset	administratively	down	
NorthOfficeRouter#						

Configuring Routing Protocols

You can use the following commands to configure routing protocols:

<code>ip route <network-number> <network-mask> {<ip-address> <exit-interface>}</code>	Sets a static route in the IP routing table
<code>ip route 0.0.0.0 0.0.0.0 {next-hop-ip exit-interface}</code>	Configures a default route
<code>no ip route {network} {mask} {next-hop-ip exit-interface}</code>	Removes a route
<code>router rip</code>	Enables a Routing Information Protocol (RIP) routing process, which places you in Router Configuration mode
<code>no auto-summary</code>	Disables automatic summarization
<code>version 2</code>	Configures the software to receive and send only RIP version 2 packets
<code>network ip-address</code>	Associates a network with a RIP routing process
<code>passive-interface interface</code>	Sets the specified interface to passive RIP mode, which means RIP routing updates are accepted by, but not sent out of, the interface
<code>show ip rip database</code>	Displays the contents of the RIP routing database
<code>default-information originate</code>	Generates a default route into RIP

The screenshot below shows the configuration of one static route and the configuration of a default route for all other networks:

```
Router (config) #ip route 172.16.5.0 255.255.255.0 192.168.1.254
Router (config) #ip route 0.0.0.0 0.0.0.0 192.168.2.254
Router (config) #show ip route
```

Configuring Network Address Translation (NAT)

Use the following commands to configure NAT, which allows private IP addresses on a local network to be translated into public IP addresses before being sent over the internet:

<pre>ip nat [inside outside]</pre>	Specifies whether the NAT operation is applied to traffic entering or leaving the router's network
<pre>ip nat inside source {list{access-list-number access-list-name}} interface type number[overload]</pre>	Establishes dynamic source translation. Use the list keyword to specify an ACL to identify the traffic that will be subject to NAT. The overload option enables the router to use one global address for many local addresses.
<pre>no ip route {network} {mask} {next-hop-ip exit-interface}</pre>	Establishes a static translation between an inside local address and an inside global address

Troubleshooting and Diagnostics

Below are commands to help you troubleshoot and perform basic diagnostics:

<pre>ping {hostname system-address} [source source-address]</pre>	Reveals basic network connectivity
<pre>traceroute {hostname system-address} [source source-address]</pre>	Traces the route that packets take to reach a destination
<pre>show interfaces</pre>	Displays detailed information about interface status, settings and counters
<pre>show ip route</pre>	Shows the routing table of the device.
<pre>show interface status</pre>	Displays the interface line status
<pre>show interfaces trunk</pre>	Lists information about the currently operational trunks and the VLANs supported by those trunks
<pre>show version</pre>	Displays information about the IOS version, uptime and hardware configuration
<pre>show running-config</pre>	Displays the current active configuration on the device

<code>show tech-support</code>	Generates a comprehensive report of the device's configuration and status (useful for advanced troubleshooting)
<code>show cdp</code>	Shows whether CDP is enabled globally
<code>show cdp neighbors [detail]</code>	Lists summary (or detailed) information about each neighbor connected to the device
<code>cdp run</code> <code>no cdp run</code>	Enables or disables Cisco Discovery Protocol (CDP) for the device
<code>show mac address-table</code>	Displays the MAC address table
<code>show vtp status</code>	Lists the current VLAN Trunk Protocol (VTP) status, including the current mode

Security Configuration

You can configure access lists to restrict and allow traffic to and from your Cisco device. Use the following commands:

<code>password <pass-value></code>	Lists the password that is required if the login command (with no other parameters) is configured
<code>username name password <pass-value></code>	Defines one of possibly multiple user names and associated passwords used for user authentication. It is used when the login local line configuration command has been used.
<code>enable password <pass-value></code>	Defines the password required when using the enable command
<code>enable secret <pass-value></code>	Sets the password required for any user to enter enable mode
<code>service password-encryption</code>	Directs the Cisco IOS software to encrypt the passwords, CHAP secrets and similar data saved in its configuration file
<code>ip access-list {standard extended} {acl-name acl-number}</code>	Creates a standard or extended ACL
<code>show version</code>	Displays information about the IOS version, uptime and hardware configuration
<code>show running-config</code>	Displays the current active configuration on the device

<code>deny source <source-wildcard></code>	Adds deny rules for an Extended ACL
<code>ip access-group {acl-name acl-number} {in out}</code>	Applies an ACL to an interface
<code>show access-lists [acl-name acl-number]</code>	Displays ACL configuration
<code>no ip access-list {standard extended} {acl-name acl-number}</code>	Removes an ACL
<code>ip domain-name name</code>	Configures a DNS domain name
<code>crypto key generate rsa</code>	Creates and stores (in a hidden location in flash memory) the keys that are required by SSH
<code>transport input {telnet ssh}</code>	Defines whether Telnet or SSH access is allowed into this switch. Both values can be specified in a single command to allow both Telnet and SSH access, which are the default settings.
<code>ntp peer <ip-address></code>	Configures the software clock to synchronize a peer or to be synchronized by a peer

The screenshot below shows commands for an extended ACL:

```

NorthOfficeRouter (config) taccess-list 25 deny 152.168.1.0 0.0.0.255
NorthOfficeRouter (config) access-list 25 permit any
NorthOfficeRouter (config) tend
NorthOfficeRoutert
¥SYS-5-CONFIG_I: Configured from console by console

NorthOfficeRoutertshow access-lists 25
Standard IP access list 25

    deny 192.168.1.0 0.0.0.255
    permit any

NorthOfficeRoutert

```

Configuring SSH and Remote Access

Use the following commands to configure SSH and remote access:

<code>hostname <name></code>	Sets a hostname (if not already configured)
<code>ip domain-name [domain-name]</code> Configures an IP domain name	Configures an IP domain name
<code>crypto key generate rsa</code>	Generates an RSA key pair for SSH
<code>ip ssh version 2</code>	Configures SSH version 2
<code>username [username] privilege [level] secret [password]</code>	Creates a local user account
<code>Router(config)# line vty [line-range]</code> <code>Router(config-line)# transport input ssh</code> <code>Router(config-line)# login local</code>	Configures VTY lines for SSH access

The screenshot below shows the generation of the RSA keys:

```

NorthOfficeRouter#config t
Enter configuration commands, one per line. End with CNTL/Z.
NorthOfficeRouter (config) #ip domain-name Cisco
NorthOfficeRouter (config) #crypto key generate rsa
The name for the keys will be: NorthOfficeRouter.Cisco
Choose the size of the key modulus in the range of 3€0 to 4086 for your
    General Purpose Keys. Choosing a key modulus greater than 512 may take
    a few minutes.

How many bits in the modulus [S12]: 2048
% Generating 2048 bit RSA keys, keys will be non-exportable... [0K]

NorthOfficeRouter (confia) #
  
```

Implementing Port Security

Use the following commands to implement port security:

<code>switchport port-security</code>	Enables port security on the interface
<code>switchport port-security maximum <number></code>	Sets the maximum number of secure MAC addresses on the port
<code>switchport port-security mac-address {mac-addr {sticky [mac-addr]}}</code>	Adds a MAC address to the list of secure MAC addresses and optionally configures them as sticky on the interface
<code>switchport port-security violation {shutdown restrict protect}</code>	Sets the action to be taken when a security violation is detected
<code>show port security [interface interface-id]</code>	Displays information about security options configured on the interface

The screenshot below shows the process of configuring port security on a switch port.

```
Switch (config) #interface FastEthernet0/7
Switch (config-if)#switchport mode access
Switch (config-if) #switchport port-security
Switch (config-if)#switchport port-security mac-address sticky
Switch (config-if)#
```

Managing User Accounts

You can use the following commands to manage user accounts:

<code>username <username> privilege <level> secret <password></code>	Creates a local user account
<code>show users</code>	Displays current user sessions
<code>no username <username></code>	Removes a user account

DHCP Configuration

Use the following commands to configure DHCP:

<code>ip address dhcp</code>	Acquires an IP address on an interface via DHCP
<code>ip dhcp pool <pool-name></code>	Configures a DHCP address pool on a DHCP server and enters DHCP Pool Configuration mode
<code>domain-name <domain></code>	Specifies the domain name for a DHCP client
<code>network network-number [mask]</code>	Configures the network number and mask for a DHCP address pool primary or secondary subnet on a Cisco IOS DHCP server
<code>ip dhcp excluded-address ip-address [last-ip-address]</code>	Specifies IP addresses that a DHCP server should not assign to DHCP clients
<code>ip helper-address address</code>	Enables forwarding of UDP broadcasts, including BOOTP, received on an interface
<code>default-router address[address2 ... address8]</code>	Specifies the default gateway for a DHCP client

The screenshot below shows the process of configuring port security on a switch port.

```

NorthOfficeRouter#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
NorthOfficeRouter(config)#ip dhcp pool Employees
NorthOfficeRouter (dhcp-config) <network 192.168.10.0 255.285.285.0
NorthOfficeRouter (dhcp-config)# default-router 192.168.10.254
NorthOfficeRouter (dhcp-config) #dns-server 192.168.1.100
NorthOfficeRouter (dhcp-config) 4

```

Monitoring and Logging

The following commands are useful for monitoring and logging:

<code>logging on</code>	Enables logging globally
<code>logging host {ip-address hostname}</code>	Configures logging to a syslog server
<code>logging trap level</code>	Sets the logging severity level
<code>terminal monitor</code>	Sends a copy of all syslog messages, including debug messages, to the Telnet or SSH user who issues this command
<code>snmp-server community <community-string> [RO RW]</code>	Enables SNMP
<code>snmp-server location <location-string></code>	Configures the SNMP server location
<code>snmp-server enable traps</code>	Enables SNMP traps

Backup, Restore and Upgrade

Use the following commands are for backup, restores and upgrades:

<code>copy running-config startup-config</code>	Saves the running configuration to startup configuration
<code>copy running-config tftp</code>	Copies the running configuration to a TFTP server
<code>copy startup-config tftp</code>	Copies the startup configuration to a TFTP server
<code>copy tftp: running-config</code>	Copies the configuration from a TFTP server to the device
<code>copy running-config flash:<file name></code>	Copies the configuration to flash
<code>copy {ftp: flash:}</code>	Copies a new IOS image to the device using TFTP or FTP

The screenshot below shows the running config being backed up to the startup config.

```
NorthOfficeRouter#copy running-config startup-config
Destination filename [startup-config]?
Building configuration.
[OK]
NorthOfficeRouter#
```

Configuring Authentication, Authorization and Accounting

The following commands are used to configure authentication, authorization and accounting (AAA):

<code>aaa new-model</code>	Enables AAA
<code>radius-server host {ip-address hostname} [auth-port port-number] [acct-port port-number] [timeout seconds] [retransmit retries] [key string]</code>	Configures the RADIUS server
<code>radius-server key {0 string 7 string string}</code>	Sets the RADIUS key
<code>aaa authentication login {default list-name} method1</code>	Configures AAA authentication
<code>aaa authorization {network exec commands level reverse-access configuration} {default list-name} method1</code>	Configures AAA authorization
<code>aaa accounting {system network exec connection commands level} {default list-name} {start-stop stop-only none} [method1]</code>	Configures AAA accounting

Common Use Cases

A common use case for Cisco switches is network segmentation and quality control through VLANs. By creating separate VLANs for wireless, phones, cameras and printers, you segment and can prioritize different traffic types (e.g., prioritizing voice over printing).

Cisco routers also enable network segmentation using standard or extended access control lists. These can restrict traffic from specific sources or limit certain traffic types entering or exiting router segments.

Here is a step-by-step guide for configuring a Cisco router for a network segment:

1. Configure the router hostname and enable a secret password.
2. Assign IP addresses to the router interfaces.
3. Either configure static routing information or configure a routing protocol to route traffic for the attached network segments.
4. Configure DHCP to distribute IP address and other DHCP options to clients that connect to the network.
5. Configure ACLs to shore up security and restrict incoming and outgoing traffic.
6. Configure NAT for internet access.
7. Enable logging.
8. Save the configuration.

Common Problem Scenarios and Troubleshooting Tips

Here are some common issues for Cisco routers and switches and tips for resolving them:

Scenario: Devices unable to communicate across VLANs or subnets

- Check VLAN configurations and trunking on switches.
- Verify the IP addresses and subnet masks.
- Check the routing table and ensure routes exist.
- Test connectivity with **ping** and **tracert**.
- Verify that ACLs are not blocking traffic.

Scenario: Interface is down or flapping

- Check physical connections and cable integrity.
- Verify the interface configuration with **show interface**.
- Shut down and re-enable the interface.
- Test different speed and duplex settings.

Scenario: Unauthorized access attempts or suspicious traffic

- Review logs and use **show logging**.
- Check ACL configurations and hit counts.
- Verify AAA and TACACS+ settings.
- Implement port security on switches.

Bonus Tips

The following tips can help you manage your Cisco devices more efficiently:

- If you aren't familiar with Cisco IOS, be sure to take advantage of the context-sensitive help — simply type ? at any point in a command to get suggestions and available
- Save time with command shortcuts and tab completion. For example, type **sh run** instead of **show running-config**.
- Use the up and down arrows (or Ctrl+P and Ctrl+N) to quickly access recent commands.
- If you don't have access to a Cisco router, there are multiple Cisco device simulators available for download that will allow you to practice and become familiar with the commands.

Always make security a top priority:

- Limit who can access your Cisco devices.
- Ensure that all accounts use encrypted long passwords.
- Enforce the principle of least privilege when assigning roles.
- Create access control lists to restrict different traffic origins and types.
- Enable logging and use a third-party monitoring solution to get alerted if your configurations are altered, either accidentally or maliciously.

How Netwrix Can Help

Regardless of who makes your routers and switches, these devices are the backbone of your network. Even a small incorrect or unauthorized configuration change can have serious consequences for your business operations. [Netwrix Auditor for Network Devices](#) provides complete visibility into your Cisco, Fortinet, Palo Alto, SonicWall, Juniper, Cisco Meraki or HPE Aruba environments.

This Netwrix solution can help you ensure that your network environment operates in optimized manner through:

Configuration change monitoring

- Tracks and reports on all changes made to device configurations
- Provides details on what was changed, when and by whom
- Provides alerting on device configuration changes that could negatively affect your network configuration

Logon attempt monitoring

- Monitors both successful and failed logon attempts to network devices
- Tracks VPN logons and remote access attempts
- Helps detect suspicious logon activity like brute force attacks

Hardware issue detection

- Reports on hardware malfunctions and failures in near real time
- Provides details on which device is affected and potential causes
- Enables proactive troubleshooting of hardware issues before they impact the network

These are just some of the many capabilities available in Netwrix Auditor for Network Devices that can empower your IT team to continuously monitor your network devices, detect threats to your perimeter security and improve overall data security in your organization.

FAQ

How can I configure a Cisco switch from the command line?

To configure a Cisco switch via command line:

1. Connect to the switch console port.
2. Enter Privileged EXEC mode with the **enable** command.
3. Enter Global Configuration mode with the **configure terminal** command.
4. Configure the hostname, IP address, VLANs and other settings using the appropriate commands.
5. Save the configuration with the command **copy running-config startup-config**.
6. Exit Global Configuration mode with either **end** or **exit**.

How can I show all interfaces on a Cisco switch?

The following command will display a summary of all interfaces on the switch, including their status, VLAN assignment, duplex mode, speed and type:

```
show interfaces status
```

What is the Cisco command line interface?

The Cisco command line interface (CLI) is a text-based interface used to configure and manage Cisco network devices like routers and switches. It allows administrators to enter commands to view device status, modify configurations and troubleshoot issues. The CLI can be accessed through a console connection or remote protocols like SSH. It uses a hierarchical mode structure, with different command modes providing access to specific configuration options.

What is the Cisco command to list all ports?

To list all ports on a Cisco switch, use this command:

```
show interfaces status
```

This will display a summary of all physical interfaces, including their status, VLAN assignment and other basic information.

What command is used to check the Cisco model?

To display information about a device, including the model number, use this command:

```
show version
```

How do I show the configuration of a Cisco device?

There are several options for viewing the configuration of a Cisco device:

- show running-config displays the current active configuration on the device.
- show startup-config displays the current saved configuration on the device.
- show run brief shows the configuration for a specific interface or feature.

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